

Space Propulsion

Can Empty Space Itself Provide A Solution?

By H. E. Puthoff

THE LAUNCH OF A MIGHTY ROCKET is truly an awe-inspiring sight. As it strains against the twin forces of gravity and inertia, we can only marvel at the progress we have made in our attempt to throw off the shackles that bind mankind to Earth.

But contemplation of the sheer expenditure of energy in such a launch must also make us wonder whether we will ever colonize even the closest planet, let alone travel to the stars. Although various propositions to surmount the difficulties involved have been put forward, we cannot help but hope that the brute force solutions we apply today will one day be replaced by alternatives we can now only dream of.

The Potential of "Empty Space"

Surprisingly enough, there are hints that potential help may emerge quite literally out of the vacuum of so-called "empty space" itself, the very medium we wish to conquer. Quantum theory tells us that empty space is not truly empty, but rather is the seat of myriad energetic quantum processes that could have profound implications for future space travel. To understand these implications it will serve us to detour for a moment to review briefly the historical development of the scientific view of what constitutes empty space.

At the time of the Greek philosophers, Democritus argued that empty space was truly a void, otherwise there would not be room for atoms to move around. Aristotle, on the other hand, argued equally forcefully that what appeared to be empty space was in fact a plenum (a background filled with substance), for did not heat and light travel from place to place as if carried by some kind of medium?

The argument went back and forth through the centuries until finally codified by Maxwell's theory of the luminiferous ether, a plenum that carried electromagnetic waves, including light, much as water carries waves across a lake. Attempts to measure the properties of this ether, or to measure the Earth's velocity through the ether (as in the famous Michelson-Morley experiment), however, met with failure. With the rise of special relativity, which did not require reference to such an underlying substrate, Einstein in 1905 effectively banished the ether in favor of the concept that empty space constitutes a true void. Ten years later, however, Einstein's own development of the general theory of relativity,

"Imaging to the Edge of Space and Time,"
an acrylic painting by James L. Cunningham.
Artwork produced under the NASA Art Program.

ision:

with its concept of curved space and distorted geometry, forced him to reverse his stand and opt for a richly endowed plenum, under the new label *spacetime metric*.

It was the advent of modern quantum theory, however, that established the quantum vacuum, so-called empty space, as a very active place, with particles arising and disappearing, a virtual plasma, and fields continuously fluctuating about their zero baseline values. Such processes are called zero-point fluctuations (ZPF) of the vacuum, reflecting the fact that such activity remains even at the zero point of temperature (absolute zero) after all thermal effects have frozen out.

Empty Space as an Energy Reservoir

What does all this have to do with space travel? At its most fundamental level, we now recognize that the quantum vacuum is an enormous reservoir of untapped energy, with energy densities conservatively estimated by Nobelist Feynman and others to be on the order of nuclear energy densities or greater. Therefore, the question is, can the ZPF energy be "mined" for practical use? If so, it would constitute a virtually ubiquitous energy supply, a veritable "Holy Grail" energy source for space propulsion.

As utopian as such a possibility may seem, researcher Robert Forward at Hughes Research Laboratories, Malibu, CA, demonstrated proof-of-principle in a paper published in *Physical Review B* in 1984, "Extracting Electrical Energy from the Vacuum by Cohesion of Charged Foliated Conductors." Forward's approach exploited a phenomenon called the *Casimir Effect*, an attractive quantum force between closely spaced metal plates, named for its discoverer, H.G.B. Casimir of Philips Laboratories in the Netherlands. The Casimir force derives from partial shielding of the interior region of the

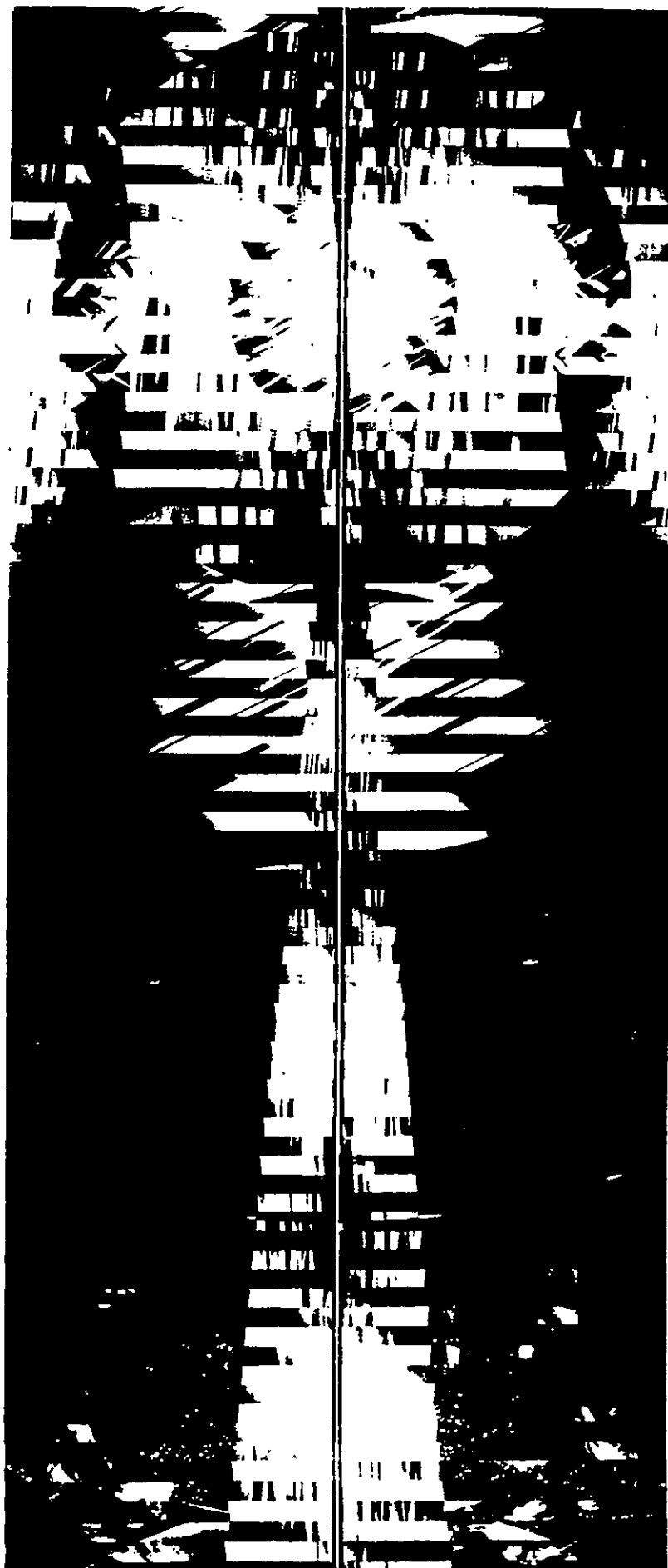


Illustration: NASA

"Flight, Time, Space,"
 an acrylic triptych by James L. Cunningham.
 Artwork created under the NASA Art Program.

plates from the background zero-point fluctuations (ZPF) of the vacuum electromagnetic field. This shielding results in the plates being pushed together by the unbalanced ZPF radiation pressures, with a consequent conversion of vacuum energy to some other form such as heat. Proof that such a process violates neither energy nor thermodynamic constraints can be found in a paper by D. Cole and myself published in *Physical Review E* in 1993, under the title "Extracting Energy and Heat from the Vacuum." Attempts to harness the Casimir Effect for vacuum energy conversion are ongoing in our laboratory and elsewhere, with formats ranging from pinch effects in plasmas to bubble collapse in turbulent fluids as in sonoluminescence.

Yet another example in which Nature herself may have taken advantage of energetic vacuum effects is discussed in a model published by ZPF researchers A. Rueda of California State University at Long Beach, B. Haisch of Lockheed, and D. Cole of IBM. In a paper published in the *Astrophysical Journal* in 1995, they propose that the vast reaches of outer space constitute an ideal environment for ZPF acceleration of nuclei and thus provide a mechanism for "powering up" cosmic rays. Details of the model would appear to account for other observed phenomena as well, such as the formation of cosmic voids. Of interest here is a proposal put forward in a report published by the U. S. Air Force (to be described later) to investigate the possibility of utilizing a "sub-cosmic ray" approach to accelerate protons in a cryogenically cooled, collision-free vacuum trap and thus extract energy from the vacuum fluctuations by this mechanism.

Origins of Gravity and Inertia

Let us now go deeper, however. What of the fundamental forces of gravity and inertia we seek to overcome in space travel? We have phenomenological theories that describe their effects (Newton's Laws and their relativistic generalizations), but what of their origins?

The first hint that these phenomena might themselves be traceable to roots in the underlying fluctuations of the vacuum came in a 1967 study published by the well-known Russian physicist Andrei Sakharov. Searching to derive Einstein's phenomenological equations for general relativity from a more fundamental set of assumptions, Sakharov came to the conclusion that the entire panoply of general relativistic phenomena could be seen as induced effects brought about by changes in the quantum-fluctuation energy of the vacuum due to the presence of matter. In this view the attractive gravitational force is more akin to the induced Casimir force discussed above, than to the fundamental inverse square law force between charged particles with which it is often compared. Although speculative when first introduced by Sakharov, this hypothesis has led to a rich and ongoing literature (including a contribution of my own in a

1989 *Physical Review A* publication) on quantum-fluctuation-induced gravity, a literature that continues to yield deep insight into the role played by vacuum forces.

Given an apparent deep connection between gravity and the zero-point fluctuations of the vacuum, it was only a matter of time before a similar connection had to be made between these self-same vacuum fluctuations and inertia. Why? It is an empirical fact that the gravitational and inertial masses have the same value, even though the underlying phenomena are quite disparate. Why, for example, should a measure of the resistance of a body to being accelerated, even if far from any gravitational field, have the same value that is associated with the gravitational attraction between bodies? Indeed, if one is determined by vacuum fluctuations, so must the other.

To get to the heart of inertia, let us consider a specific example. You are standing on a train in the station. As the train leaves the platform with a jerk, you could be thrown to the floor. What is this force that knocks you down, seemingly





Illustration: MSA

coming out of nowhere? This phenomena, which we conveniently label inertia and go on about our physics, is a subtle feature of the universe that has perplexed generations of physicists from Newton to Einstein. Since in this example the sudden disquieting imbalance results from acceleration "relative to the fixed stars," in its most provocative form one could say that it was the "stars" that delivered the punch. This key feature was emphasized by the Austrian philosopher of science Ernst Mach, and is now known as Mach's Principle. Nonetheless, the mechanism by which the stars might do this deed has eluded convincing explication—until now.

Addressing this issue in a paper entitled "Inertia as a Zero-Point Field Lorentz Force," published in *Physical Review A* in 1994, I and my colleagues Haisch and Rueda (mentioned earlier) were successful in tracing the problem of inertia and its connection to Mach's Principle to the ZPF properties of the vacuum. In a sentence, although a uniformly moving body does not experience a drag force from the (Lorentz-invariant) vacuum fluctuations, an *accelerated*

body meets a resistance (force) proportional to the acceleration. By *accelerated* we mean, of course, accelerated relative to the fixed stars. It turns out that an argument can be made that the quantum fluctuations of distant matter structure the local vacuum-fluctuation frame of reference. Thus, in the example of the train the punch was delivered by the wall of vacuum fluctuations, acting as a proxy for the fixed stars, through which one attempted to accelerate.

Again, what does all this have to do with space travel? *There is experimental evidence that vacuum fluctuations can be altered by technological means. This leads to the corollary that, in principle, gravitational and inertial masses can also be altered.*

The evidence for the alteration of vacuum fluctuations is found in the research area called *cavity quantum electrodynamics* (QED). There, excited atoms are passed through Casimir-like cavities whose structure suppresses electromagnetic cavity modes at the transition frequency between the atom's excited and ground states. The result is that the so-

called "spontaneous" emission time is lengthened considerably (for example, by factors of ten), simply because spontaneous emission is not so spontaneous after all, but rather is driven by vacuum fluctuations. Eliminate the modes and you eliminate the zero-point fluctuations of the modes, hence suppressing decay of the excited state. As stated in an April 1993 *Scientific American* review article on cavity QED, "An excited atom that would ordinarily emit a low-frequency photon cannot do so, because there are no vacuum fluctuations to stimulate its emission...."

The Forward Report to the Air Force

Does anyone take seriously the concept that it might be possible to alter mass, with a view to easing the energy burden of future spaceships? In fact, the implication for space travel of our innovative ZPF-based inertia model has attracted the attention of the Advanced Concepts Office of the Propulsion Directorate of the Phillips Laboratory at Edwards Air Force Base. This office is charged with initiating research relevant to the development of 21st century space propulsion, and it is well understood that a fundamental understanding of inertia could well contribute to new concepts in this area. It was in this context that Robert Forward (the same Forward mentioned earlier!), a respected authority in the area of gravitation theory and measurement, accepted an assignment to review our concept and program. He recommended a broad, multi-pronged effort, involving laboratories from around the world, to investigate our inertia model experimentally.

After a one-year investigation Forward finished his study and submitted his report to the air force, who published it under the title *Mass Modification Experiment Definition Study*. The abstract reads in part:

"Many researchers see the vacuum as a central ingredient of 21st century physics. Some even believe the vacuum may be harnessed to provide a limitless supply of energy. This report summarizes an attempt to find an experiment that would test the Haisch, Rueda and Puthoff (HRP) conjecture that the mass and inertia of a body are induced effects brought about by changes in the quantum-fluctuation energy of the vacuum.... It was possible to find an experiment that might be able to prove or disprove that the inertial mass of a body can be altered by making changes in the vacuum surrounding the body."

With regard to action items, Forward in fact recommends a ranked list of not one but *four* experiments to be carried out to address the ZPF-inertia concept and its broad implications, including investigation of the proposed subcosmic-ray energy device mentioned earlier.

Warp Drives and Wormholes

Since we are pushing the frontiers, we might as well address perhaps one of the most speculative, but nonetheless scientifically grounded, proposals of all: the Alcubierre Warp Drive. Taking on the challenge of determining whether Warp Drive à la Star Trek was a scientific possibility, general relativity theorist Miguel Alcubierre of the University of Wales set himself the task of determining

whether faster-than-light travel was possible within the constraints of standard theory. Although this clearly could not be the case in the flat space of special relativity, general relativity permits consideration of altered spacetime metrics where such a possibility is not *a priori* ruled out. Alcubierre's further self-imposed constraints on an acceptable solution included the requirements that no net time distortion should occur (breakfast on Earth, lunch on Alpha Centauri, and home for dinner with your wife and children, not your great-great-great grandchildren), and that the occupants of the spaceship were not to be flattened against the bulkhead by unconscionable accelerations.

A solution meeting all of the above requirements was found and published by Alcubierre in *Classical and Quantum Gravity* in 1994. The solution discovered by Alcubierre involved the creation of a local distortion of spacetime such that spacetime is expanded behind the spaceship, contracted ahead of it, and yields a hypersurfer-like motion faster than the speed of light as seen by observers outside the disturbed region. In essence, on the outgoing leg of its journey the spaceship is pushed away from Earth and pulled toward its distant destination by the engineered local expansion of spacetime itself. (For follow-up on the broader aspects of "metric engineering" concepts, one can refer to a paper published by myself in *Physics Essays* in 1996.) Interestingly enough, the engineering requirements rely on the generation of macroscopic, negative-energy-density, Casimir-like states in the quantum vacuum of the type discussed earlier. Unfortunately, meeting such requirements is presently beyond our technological reach.

Finally, of course, it has been known for some time that general relativity permits the possibility of *wormholes*, topological tunnels which in principle could connect distant parts of the universe, a cosmic subway so to speak. Publishing in the *American Journal of Physics* in 1988, theorists Morris and Thorne have outlined the requirements for traversable wormholes, and have found that, in principle, the possibility exists provided one has access to (yes, you guessed it) engineerable, Casimir-like, negative-energy-density quantum vacuum states.

"Magic"

Where does all this leave us? As we peer with longing into the heavens from the depth of our gravity well, hoping for some "magic" solution that will launch our spacefarers first to the planets and then to the stars, we are reminded of Arthur C. Clarke's phrase that highly advanced technology is essentially indistinguishable from magic. Fortunately, such magic appears to be waiting in the wings of our deepening understanding of the quantum universe in which we live, and it is only a matter of time before such magic will become the handmaiden of mankind's drive to explore the beckoning highways and byways of interstellar space. *

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